

b) a) Payout

Variables: T = time of touchdown

$T \sim \text{Geometric}(p)$

$$E[T] = 1/p$$

$$\text{Var}(T) = \frac{1-p}{p^2}$$

$$W = 150 - 10(T-1)$$

$$= 150 - (10T - 10)$$

$$= 160 - 10T$$

$$E[W] = 160 - 10E[T]$$

$$= 160 - 10/p$$

Variance:

$$\text{Var}(W) = \text{Var}(160 - 10T)$$

$$= (10)^2 \text{Var}(T)$$

$$= \frac{100(1-p)}{p^2}$$

b) G = single game payout

$$P(T \leq 15) = 1 - (1-p)^{15} \rightarrow 30$$

$$P(15 < T \leq 30) = (1-p)^{15} - (1-p)^{30} \rightarrow 10$$

$$P(T > 30) = (1-p)^{30} \rightarrow 0$$

$$E[G] = 30(1 - (1-p)^{15}) + 10((1-p)^{15} - (1-p)^{30})$$

$$= 30 - 20(1-p)^{15} - 10(1-p)^{30}$$

$$E[X] = 20 \cdot E[G]$$

$$= 600 - 400(1-p)^{15} - 200(1-p)^{30}$$